



ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : BACHELOR OF COMPUTER ENGINEERING TECHNOLOGY (COMPUTER SYSTEMS) WITH HONOURS

Course Code & Name : IPB 49906 FINAL YEAR PROJECT 2

Course Lecturer(s) : Ts. DIYANA BT AB KADIR (FYP2 BCE COORDINATOR)

Submission Title : PROGRESS REPORT

Deadline : Day 22 Month MAY Year 2026 Time 11:50PM

Penalties :

- 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted.
- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

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☒	I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations.
☒	This submission is my own, unless indicated with proper referencing.
☒	This submission has not been previously submitted or published.
☒	This submission follows the requirements stated in the course.

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
22/5/2026	MD EJAJ MAHMUD	52222222123

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)
22/5/2026	IPB 49906	Progress Report 5	52222222123



**UNIVERSITI KUALA LUMPUR
RUBRIC FOR PROJECT PROGRESS**

COURSE CODE	IPB 49906	TOTAL MARKS (50 MARKS)
COURSE NAME	FINAL YEAR PROJECT 2	
COURSE SEMESTER/ YEAR	MARCH 2026	
STUDENT NAME	MD EAJ MAHMUD	
STUDENT ID	5222222123	
ASSESSMENT DATE	WEEK 1 – WEEK 11	Smart Recycling: Reward-Based Reverse Vending Machine Using Proximity Sensor Detection for Plastic Waste Management
ASSESSMENT DUE	22 MAY 2026	
NAME OF SUPERVISOR	Dr. Grad. Engr. Hannah Binti Sofian	

NO.	CRITERIA	Description	WEIGHTAGE	SCORE 1-10	ACTUAL SCORE
1	Consultation with Supervisor	Meet Supervisor with significant progress	1		
2	Timelines	Record of research progress with respect to the Timelines	1		
3	Research progress	Show progress in research with related information and resources	1		
4	Proposed Idea	Ability to apply higher order thinking	1		
5	Self-driven & Independency	Ability to perform independently	1		
TOTAL SCORE (50 MARKS)					0



PROGRESS REPORT NO. 5

Student's Name: MD EAJ MAHMUD

FYP Code : IPB49906

Semester	07	Week	11
Meeting No.	01	Date	29/05/2026

Progress (fill in by Student)

Updated Progress Report 5: Week 11 / Week 12 Late Update (30 May 2026)

By the latest update, the project has progressed from a standalone Arduino prototype into a complete IoT-enabled Smart Recycling Reverse Vending Machine system. The physical prototype still uses Arduino Mega 2560 as the real-time controller for the sensors, LCD, LEDs, buzzer, gate servo, and reward dispenser. The newly added ESP32 DevKit V1 works as a WiFi bridge that sends machine events to Cloud Firestore, allowing the web dashboard to display live machine status, accepted/rejected item counts, reward dispenser activity, telemetry logs, and diagnostic information.

Completed Physical Prototype Progress

- The front panel, angled bottle chute, mid-tube gate servo, LCD, LEDs, buzzer, power switch, Arduino Mega, buck converters, and sensor modules have been assembled and tested.
- The RVM accepts PET bottles using rule-based proximity detection and rejects metal cans using the inductive sensor. The gate servo opens only for accepted PET bottles.
- The reward mechanism has been added using two SG90 servos on D10 to release a pen after a successful PET acceptance.
- The firmware was improved to main.ino v4, which keeps the v3 sensor classification logic and adds IoT event transmission through Serial1.

IoT Integration Added Using ESP32

- An ESP32 DevKit V1 was added as the IoT communication module because the Arduino Mega does not have built-in WiFi.
- The Arduino Mega sends plain-text event words through Serial1 TX1 pin D18 at 9600 baud. The ESP32 receives the signal on RX2 GPIO16 through a resistor divider to protect the 3.3V ESP32 input.
- The ESP32 signs in to Firebase and uploads machine status, events, alerts, counts, bin state, firmware version, and heartbeat data to Cloud Firestore.
- The complete chain was tested successfully: bottle/can input -> Arduino Mega -> Serial link -> ESP32 -> Firestore -> live IoT web dashboard.
- The real test recorded acceptedCount = 2, rejectedCount = 1, and penDispensedCount = 2 on 30 May 2026.

Current IoT Architecture Summary

Layer	Implementation	Purpose
Physical RVM	Arduino Mega 2560, proximity sensors, IR, servos, LCD, LEDs, buzzer	Performs real-time detection, acceptance/rejection, reward release, and local feedback.
IoT Bridge	ESP32 DevKit V1 with UART RX2 GPIO16 and WiFi	Receives machine events from Arduino Mega and uploads telemetry to the cloud.
Cloud Backend	Firebase Cloud Firestore with device-authenticated writes	Stores machine status, events, alerts, and audit-ready telemetry records.
Web Dashboard	React/Vite IoT portal at https://rvm.ejaj.website/	Shows live/simulated telemetry, analytics, event logs, diagrams,



		diagnostics, and supervisor review mode.
Mobile App	Flutter app starter https://ejajmamud.github.io/rvm-iot/	Supports mobile monitoring and can be expanded for push notifications and QR reward features.

Updated Pin and Communication Mapping

- Inductive proximity sensor: D4; capacitive proximity sensor: D5; IR entry sensor: D11; gate servo: D9; reward servos: D10; buzzer: D8; LCD: D20/D21.
- main.ino v4 code uses Green LED on D6 and Red LED on D7. This must match the physical wiring for the final demonstration.
- Mega-to-ESP32 telemetry line: Mega TX1 D18 -> resistor divider -> ESP32 RX2 GPIO16. Common ground is required.
- Events transmitted to ESP32: PET_ACCEPTED, METAL_REJECTED, PEN_DISPENSED, BIN_FULL, and BIN_CLEARED.

IoT Web Dashboard Progress

- The web dashboard includes Overview, Live Simulator, Real-Time Telemetry, Analytics, Alerts, Hardware Pinout, Engineering Diagrams, Component Datasheets, Construction Gallery, Admin Settings, and Supervisor Review sections.
- The Engineering Documentation module includes System Architecture, Hardware Block Diagram, IoT Data Flow, Arduino State Machine, Sensor Classification Logic, Firebase Database Schema, Role-Based Security Model, and Power Distribution diagrams.
- The dashboard has role-based access, read-only supervisor/guest mode, admin PIN protection, demoGuard protection for restricted actions, machine health score, heartbeat monitor, bin forecast, reward stock monitor, event timeline, alerts, and analytics.
- Production build was verified with npm run build. The project repository is <https://github.com/ejajmamud/rvm-iot.git> and the deployed dashboard is <https://rvm.ejaj.website/>.

Known Issues and Mitigation

- The HC-SR04 ultrasonic sensor currently returns timeout in standalone testing, indicating a likely faulty sensor or power/wiring issue. This does not affect the main PET acceptance and metal rejection demonstration because the firmware treats timeout as not full.
- The bin-full code path, event word, Firestore alert path, and dashboard display are prepared. A known-good HC-SR04 sensor should be swapped in before long-term deployment.
- The ESP32 should be powered from a USB power bank during the demonstration. Laptop USB power caused noise/ground-loop behaviour that affected the capacitive sensor and created false accepts.
- The physical structure remains a final-year prototype using PVC foam board and demo-level wiring; future improvement should use stronger enclosure material, terminal blocks or PCB wiring, and long-term reliability testing.

Evidence Added

The following architecture diagram summarises the final IoT workflow implemented for Progress Report 5.

Next Immediate Action Before Product Demonstration

- Stop any dashboard simulator before real hardware testing so real events are not mixed with fake seeded transactions.
- Reset dashboard counts before the demo by powering off the ESP32, clearing Firestore counts, and then powering on the ESP32 again.
- Use the iPhone hotspot in 2.4 GHz compatibility mode for the ESP32 and verify dashboard online status before the demonstration.
- Prepare a backup screen recording of the dashboard receiving live events in case venue WiFi or hotspot is unstable.



Supervisor Feedback Included From 30 May Discussion

- Supervisor advised to start the technical paper with methodology and to clearly explain the block diagram and flow chart.
- Supervisor asked for project limitation, weakness, future recommendation, scope and limitation explanation for poster/report.
- Supervisor asked to add problems faced and solutions in Chapter 4 / results discussion, including the burnt Arduino issue and power measurements.
- Supervisor advised that AI self-sorting can be added only as a future recommendation because implementing camera-based AI/ML within the remaining time is not practical.
- Supervisor accepted that current IoT is based on apps, sensors and internet connectivity through ESP32, Firebase, dashboard and monitoring.

Problem Faced and Solution Summary

Problem Faced	Cause / Observation	Solution Applied
Arduino board burnt earlier	Smoke/overheat observed at a component and LCD stopped appearing after the issue.	Replaced the Arduino board and separated power rails using buck converters.
Power supply risk	12V adapter was too high for sensors/servos without regulation.	Measured and used two buck converters: about 7.50V for proximity sensors and 5.00V for servos.
ESP32 laptop USB noise	Laptop USB introduced noise/ground-loop behaviour and caused false PET accepts.	Powered ESP32 from a USB power bank during demo and kept common signal ground for UART reference.
HC-SR04 timeout	Standalone testing showed constant timeout, likely faulty sensor or power/wiring issue.	Core demo continues; bin-full software/cloud path prepared; sensor replacement planned.
Sensor false triggers	Capacitive/IR sensors are sensitive to wall distance and trim-pot setting.	Added clearance, adjusted sensitivity, and used a classification window to avoid false metal/PET decisions.

Updated Cost Note

The approximate total spending is now around RM500 because one Arduino board was damaged during development and had to be replaced. The final build also added ESP32 IoT hardware, power regulation parts, wiring, prototype materials, sensors, servos, and dashboard development resources.

Operational Flow Chart Added

The following flow chart is added to support the supervisor request to explain the method, block diagram, and system flow clearly.

Appendix 1: Progress Report 5 Final Prototype Evidence

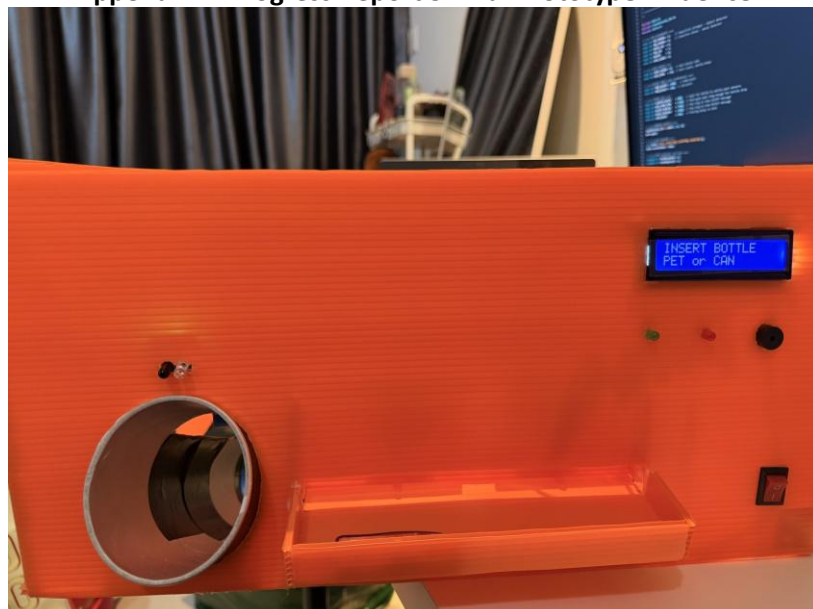


Figure A1. Front panel of the prototype with LCD user instruction displayed.

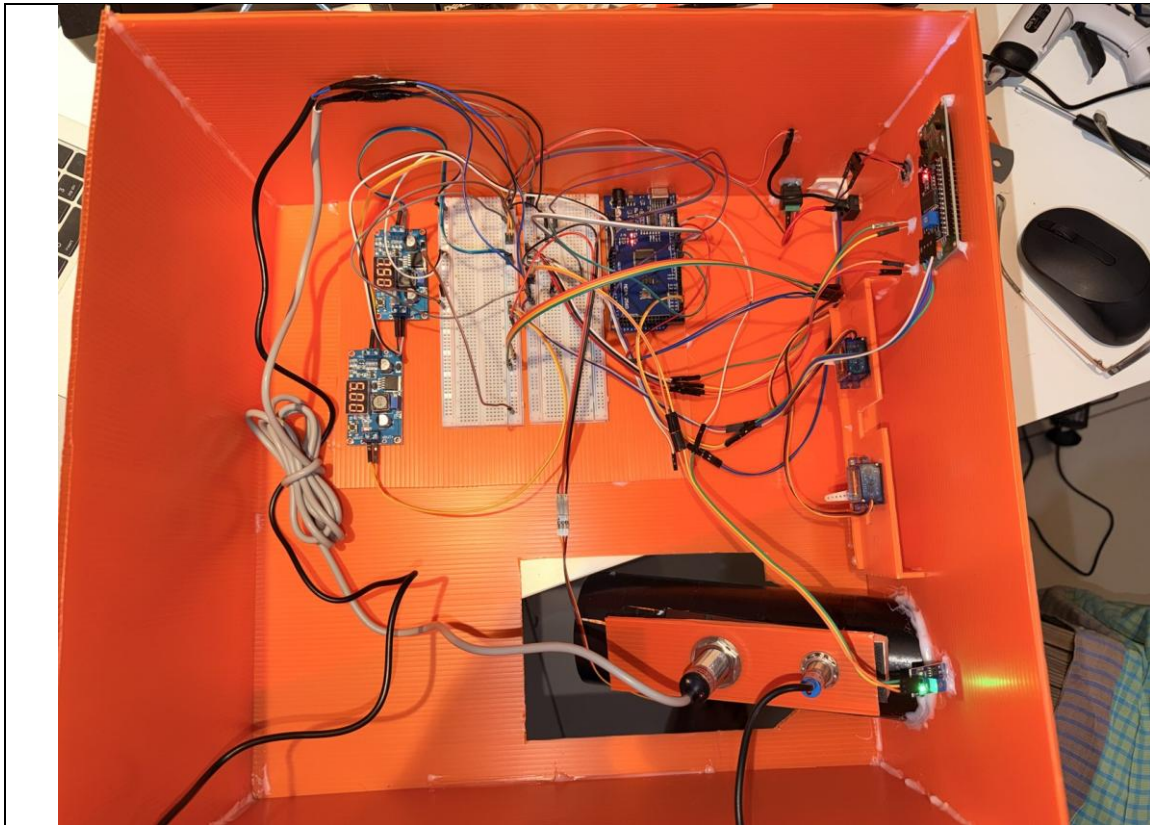


Figure A2. Internal wiring layout with Arduino Mega 2560 and connected modules.

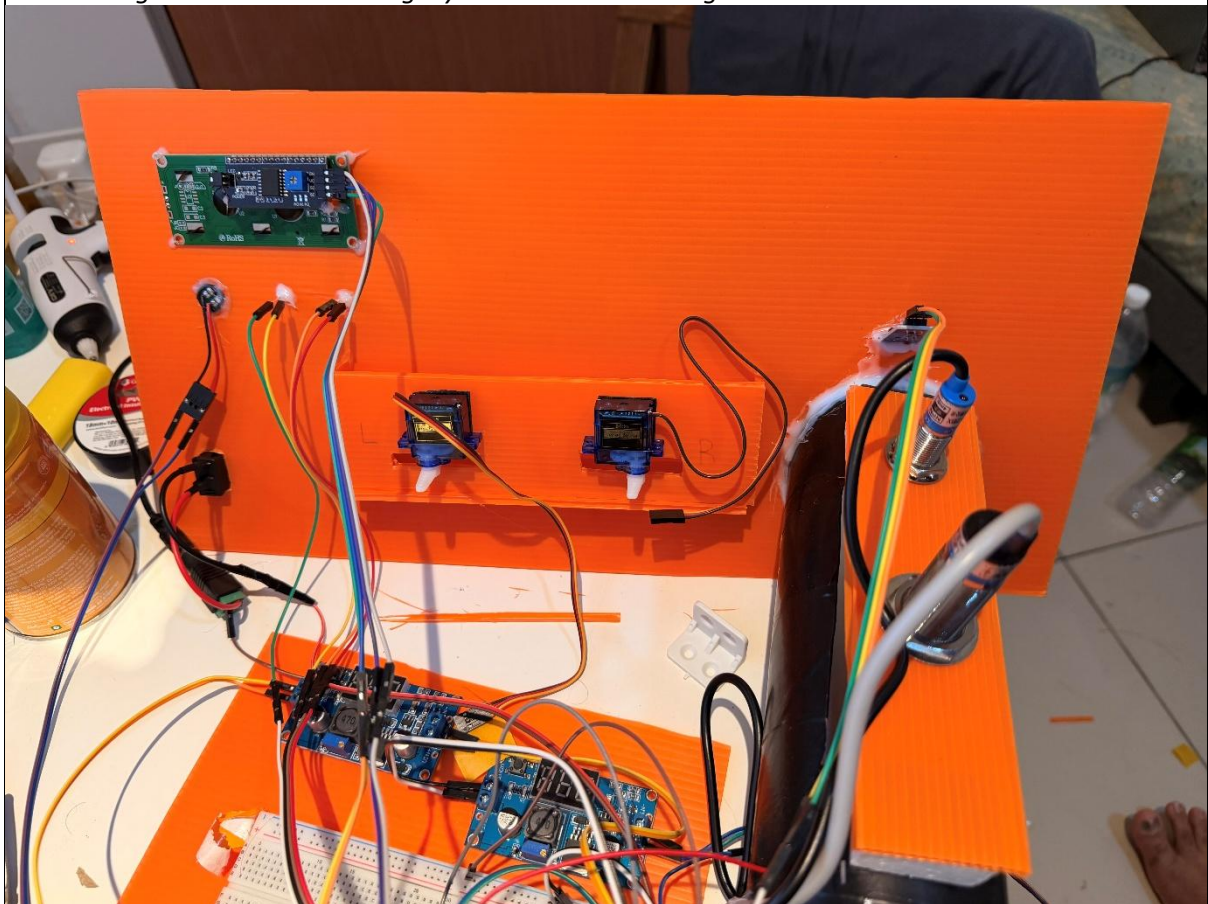
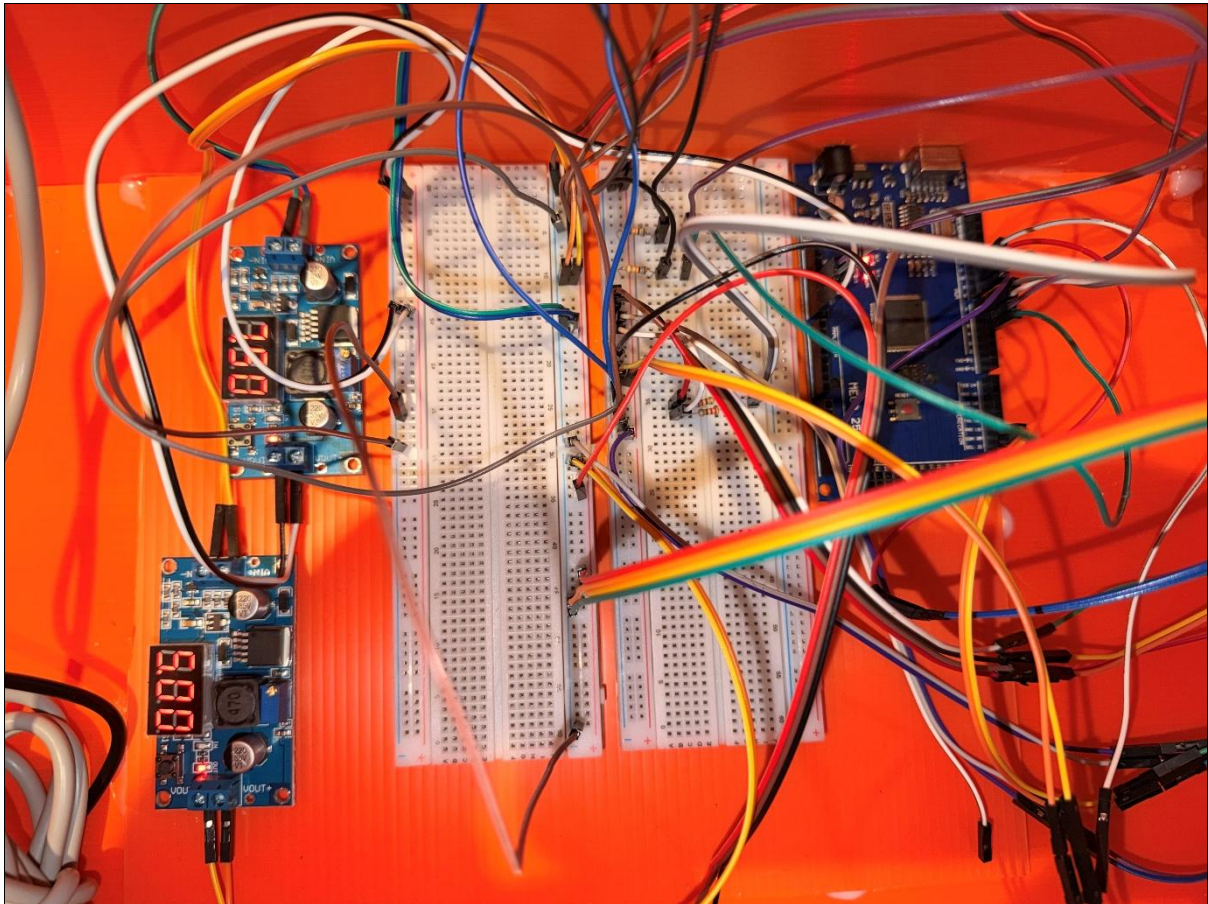


Figure A3. Completed front-panel rear view with reward pen dispenser module



. Figure A4. Top Close view of Wiring



Figure A5. Bottom view of Drop Tube with Receiver gate servo



Figure A6. Main Power Intake Socket , Arduino USB & Arduino power on right panel.

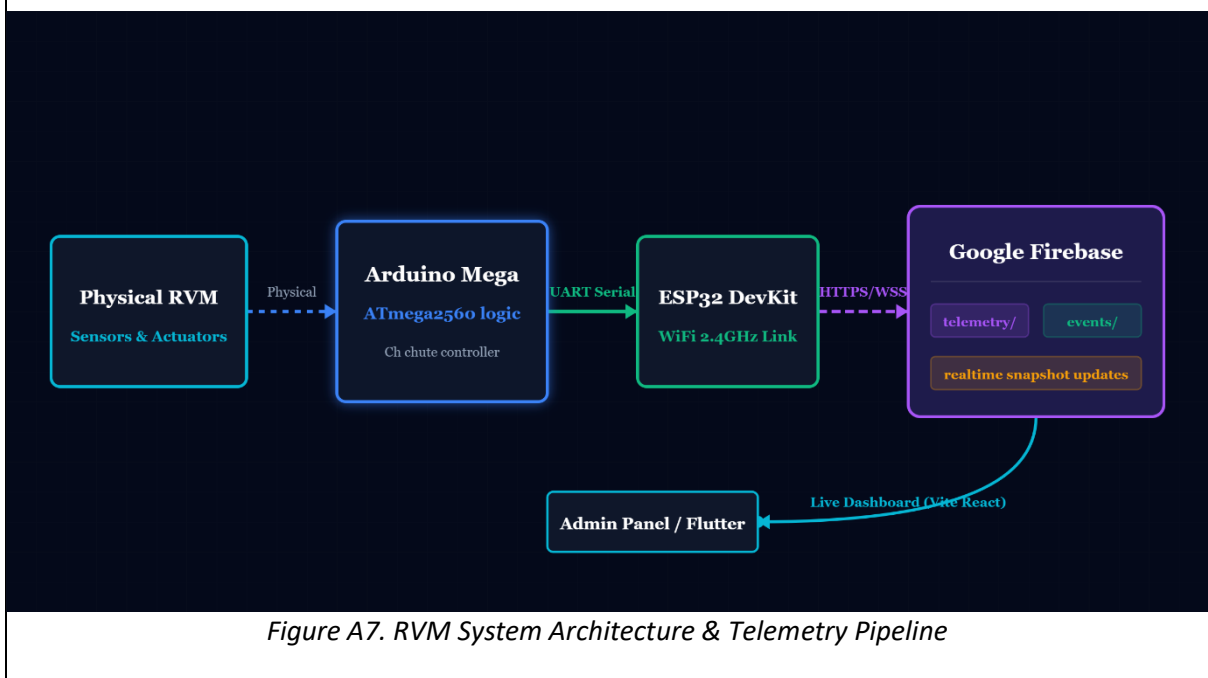


Figure A7. RVM System Architecture & Telemetry Pipeline



Figure A8. Central ATmega2560 Hardware Connection Bus

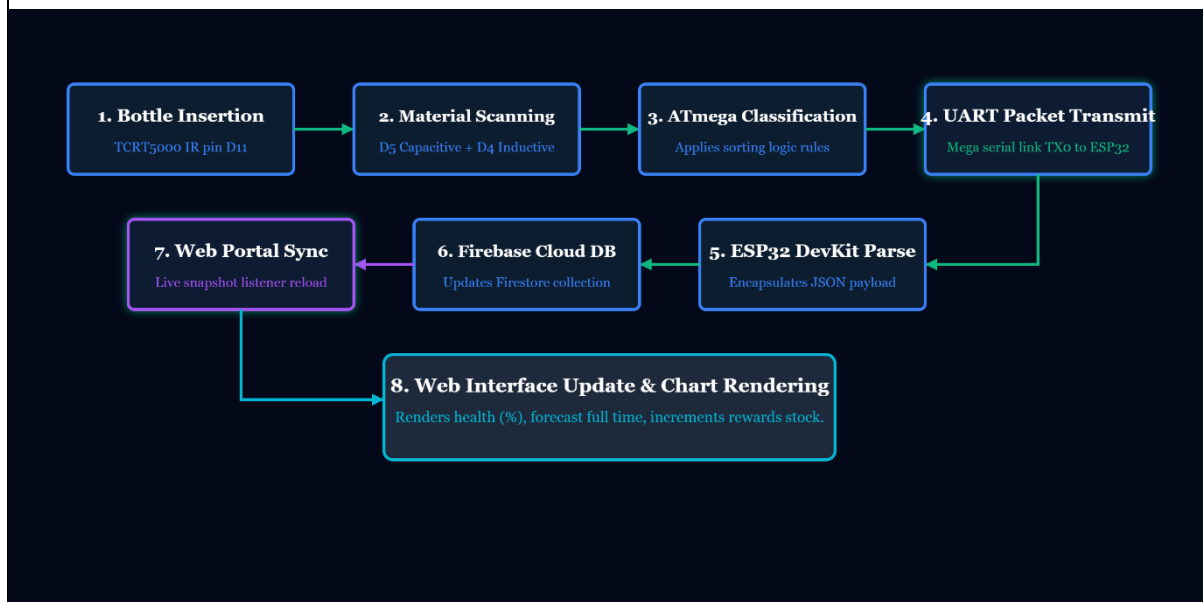


Figure A9. End-to-End IoT Data Flow & Ingestion Pipeline

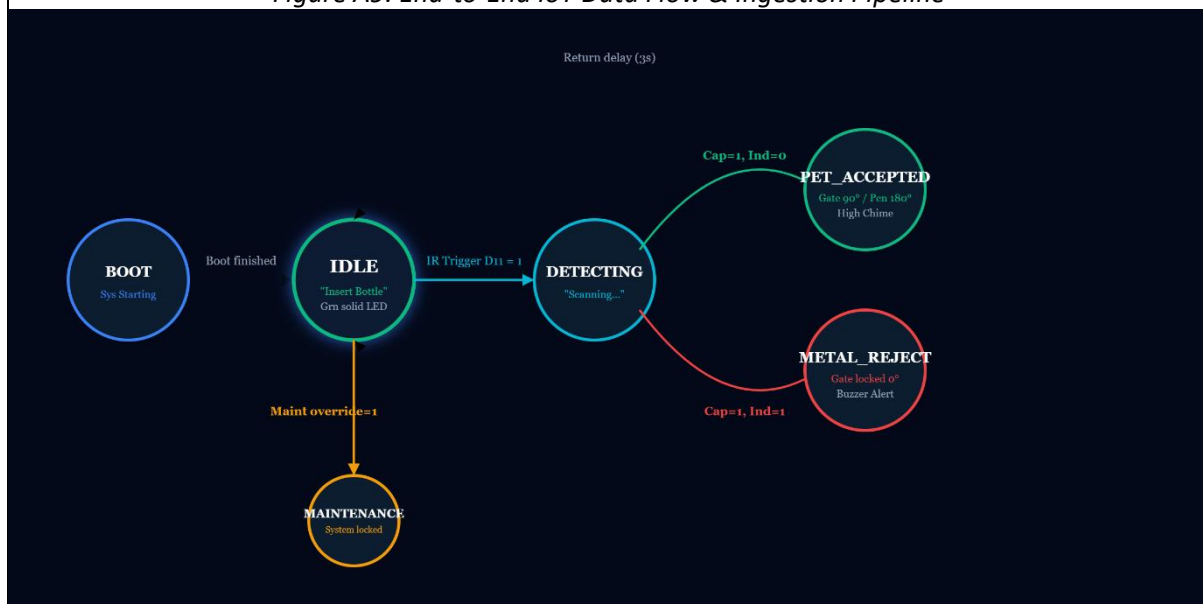


Figure A10. Arduino State Machine Behaviour Diagram

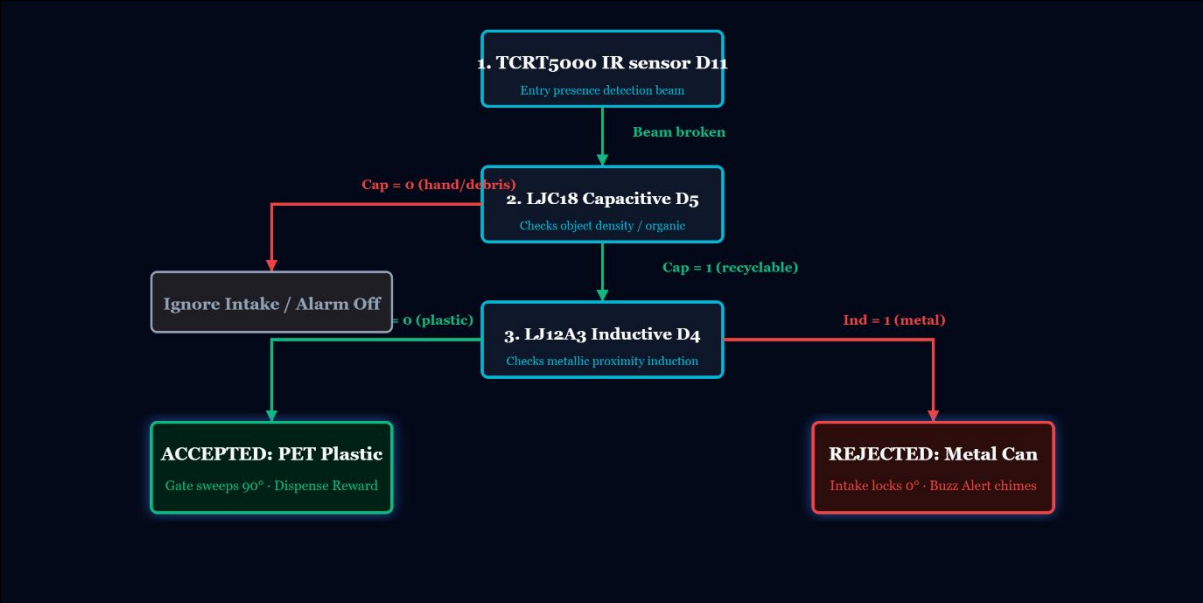


Figure A11. Rule-Based Sensor Classification Decision Tree



Figure A12. Figure A11. Rule-Based Sensor Classification Decision Tree



Comments (by Supervisor)	Signature and Name
ok good, progress 6, now submit to VLE, Report of Technical Paper. 11 progress Report + abstract, Final Report and Technical Paper.	 Dr. Hajjah Usamah binti Shams